

Laboratory Brief 1: Visual Search

Due: Thursday, September 10 at 11:59 PM Value: 55 points Format: Individual laboratory brief

Purpose

Estimate and compare search slopes for feature and conjunction searches, then decide whether the class pattern supports parallel or serial search.

Your task

- Plot mean response time against set size for both search conditions.
- Estimate each slope using $(RT \text{ at set size } 16 - RT \text{ at set size } 4) / 12$. Report milliseconds per item.
- Compare accuracy before interpreting speed; identify any speed-accuracy tradeoff.
- Write a 700-900 word brief with question, operational definitions, results, interpretation, and one design limitation.

Provided evidence or data

Set size	Feature RT (ms)	Feature accuracy	Conjunction RT (ms)	Conjunction accuracy
4	432	.98	518	.97
8	445	.98	648	.95
12	459	.97	784	.94
16	474	.97	925	.92

Submit

- PDF laboratory brief with one figure and an APA-style figure caption.
- Spreadsheet or table showing both slope calculations.

Important note

These are simulated class means for instructional use. Treat condition as the independent variable and response time and accuracy as dependent variables.

Grading criteria

Criterion	Pts	Full-credit evidence
Question and variables	8	States the question and operational definitions accurately.
Analysis	15	Calculations are correct, transparent, and tied to the supplied data.
Figure or table	10	Presents a labeled, readable display with units and an informative caption.
Interpretation	15	Explains the pattern using the relevant theory without overstating causality.
Limitations and form	7	Identifies a meaningful limitation and follows the brief format.

Submission standards

Upload a readable PDF unless the handout specifies another format. Include your name, course code, and assignment title. Cite borrowed ideas and retain the editable source file. The learning portal timestamp is the official submission time.